Sample Paper 15

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

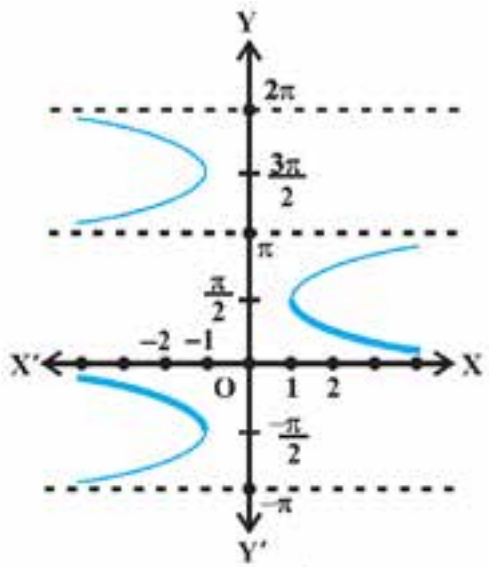
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the domain of the function $y = \text{cosec}^{-1}x$ as shown in the graph?



(a) $(-\infty,\infty)$ (b) $(-1,1)$

(c) $(-\infty,-1] \cup [1,\infty)$ (d) $[0,\pi]$

2. If $\sin(x+y) = \log(x+y)$, then dy/dx is equal to

(a) 2 (b) -2

(c) 1 (d) -1

3. The area bounded by the parabola $y^2 = 8x$ and its latusrectum is

(a) 16/3 sq units (b) 32/3 sq units

(c) 8/3 sq units (d) 64/3 sq units

4. Evaluate $\int \frac{x^3 - x^2 + x - 1}{x-1} dx$

(a) $\frac{x^3}{3} + x + C$ (b) $\frac{x^3}{4} + x + C$

(c) $\frac{x^3}{5} + x + C$ (d) $\frac{x^3}{6} + x + C$

5. If $\vec{a}$ and $\vec{b}$ are two unit vectors inclined at an angle $\pi/3$, then the value of $|\vec{a} + \vec{b}|$ is

(a) equal to (b) greater than 1

(c) equal to (d) less than 1

6. The point of intersection of lines

$$\frac{x-5}{3} = \frac{y-7}{-1} = \frac{z+2}{1}$$

and $\frac{x+3}{-36} = \frac{y-3}{2} = \frac{z-6}{4}$ is

(a) $(5,7,-2)$ (b) $(-3,3,6)$

(c) $(2,10,4)$ (d) $\left(21,\frac{5}{3},\frac{10}{3}\right)$

7. The area bounded by $y = \log x$, X-axis and ordinates $x = 1$, $x = 2$ is

(a) $\frac{1}{2}(\log 2)^2$ (b) $\log(2/e)$

(c) $\log(4/e)$ (d) $\log 4$

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8. It is given that the events A and B are such that $P(A)=\frac{1}{4}$, $P(A/B)=\frac{1}{2}$ and $P(B/A)=\frac{2}{3}$. Then, $P(B)$ is equal to

(a) $\frac{1}{2}$ (b) $\frac{1}{6}$

(c) $\frac{1}{3}$ (d) $\frac{2}{3}$

9. If $\vec{a}\cdot\vec{b}=-|\vec{a}||\vec{b}|$, then the angle between a and b is

(a) 45° (b) 180°

(c) 90° (d) 60°

10. Equation of the line passing through $(2,-1,1)$ and parallel to the line $\frac{x-5}{4}=\frac{y+1}{-3}=\frac{z}{5}$ is

(a) $\frac{x-2}{4}=\frac{y+1}{-3}=\frac{z-1}{5}$

(b) $\frac{x-2}{4}=\frac{y+1}{3}=\frac{z-1}{5}$

(c) $\frac{x-2}{-4}=\frac{y+1}{-3}=\frac{z-1}{5}$

(d) None of the above

11. If A is 3×4 matrix and B is a matrix such that $A'B$ and BA' are both defined, then B is of the type

(a) 4×4 (b) 3×4

(c) 4×3 (d) 3×3

12. $\int\frac{\sec^2(\sin^{-1}x)}{\sqrt{1-x^2}}dx$ is equal to

(a) $\sin(\tan^{-1}x)+C$ (b) $\tan(\sec^{-1}x)+C$

(c) $\tan(\sin^{-1}x)+C$ (d) $-\tan(\cos^{-1}x)+C$

13. For all real values of x, the minimum value of $\frac{1-x+x^2}{1+x+x^2}$ is

(a) 0 (b) 1

(c) 3 (d) $\frac{1}{3}$

14. $\int\sqrt{1+\cos x}\,dx$ is equal to

(a) $2\sin\left(\frac{x}{2}\right)+C$

(b) $\sqrt{2}\sin\left(\frac{x}{2}\right)+C$

(c) $2\sqrt{2}\sin\left(\frac{x}{2}\right)+C$

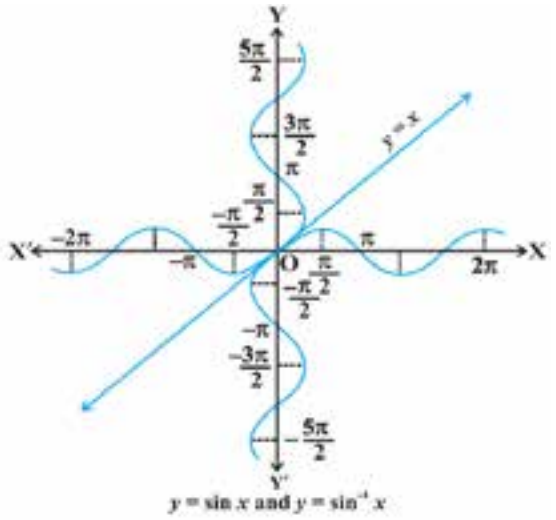
(d) $\frac{1}{2}\sin\left(\frac{x}{2}\right)+C$

15. If $A=\begin{bmatrix}3 & 1\\ -1 & 2\end{bmatrix}$, then

(a) $A^2+7A-5I=O$ (b) $A^2-7A+5I=O$

(c) $A^2+5A-7I=O$ (d) $A^2-5A+7I=O$

16. Which vertical lines act as asymptotes for the graph of $y=\tan x$?



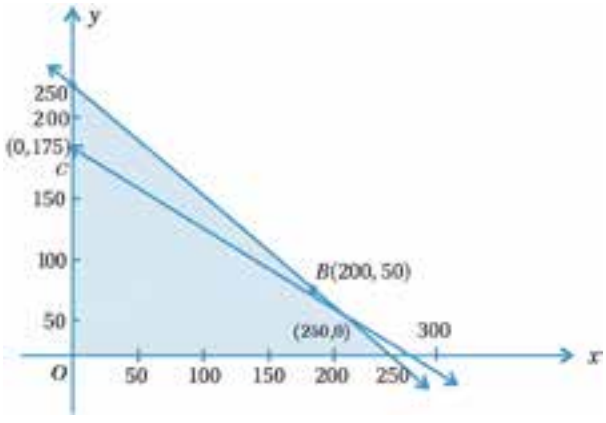
(a) $x=\pi$ and $x=2\pi$

(b) $x=-\pi/2,\pi/2,3\pi/2$

(c) $x=-\pi,0,\pi$

(d) $x=-\pi/4,\pi/4,3\pi/4$

17. The corner points of the bounded feasible region of an LPP are O(0,0), A(250,0), B(200,50), and C(0,175). If the maximum value of the objective function $Z=2ax+$ by occurs at the points A(250,0) and B(200,50), then the relation between a and b is to be determined.



(a) 2a=b (b) 2a=3b

(c) a=b (d) a=2b

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18. The maximum value of xe^{-x} is

(a) e (b) $1/e$

(c) $-e$ (d) $-1/e$

19. **Assertion :** The general solution of $\frac{dy}{dx}+y=1$ is $ye^{x}=e^{x}+c$

Reason : The number of arbitrary constants is in the general solution of the differential equation is equal to the order of differential equation.

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

20. **Assertion :** The elimination of four arbitrary constants in $y=(c_1+c_2+c_3e^{c_4})x$ results into a differential equation of the first order $x\frac{dy}{dx}=y$

Reason : Elimination of n arbitrary constants requires in general a differential equation of the nth order.

(a) Assertion is true, reason is true, reason is a correct explanation for assertion.

(b) Assertion is true,reason is true, reason is not a correct explanation for assertion.

(c) Assertion is true, reason is false.

(d) Assertion is false, reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. Find the vector equation of the line passing through the point $A(1,2,-1)$ and parallel to the line $5x-25=14-7y=35z$.

OR

The x-coordinate of point on the line joining the points $P(2,2,1)$ and $Q(5,1,-2)$ is 4. Find its z -coordinate.

22. Show that the function $f(x)=x^3-3x^2+3x$, $x\in R$ is increasing on R.

23. Find the general solution of following equation $e^{x}\tan y\,dx+(1-e^{x})\sec^2 y\,dy=0$

24. If $P(not\ A)=0.7$, $P(B)=0.7$ and $P(B/A)=0.5$, then find $P(A/B)$.

25. Maximize $Z=x+y$, subject to $x-y\le -1$, $x+y\le 0$, x, $y\ge 0$.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. Show that $y=\log(1+x)-\frac{2x}{2+x}$, $x>-1$ is an increasing function of x throughout its domain.

27. Show that the modulus function $f:R\to R$, given by $f(x)=|x|$, is neither one-one nor onto, where $|x|$ is x, if x is positive or 0 and $|x|$ is $-x$, if x is negative.

OR

Show that the Signum function of $f:R\to R$, given by

$$f(x)=\begin{cases}1, & if\ \ x>0\\ 0, & if\ \ x=0\\ -1, & if\ \ x<0\end{cases}$$

is neither one-one nor onto.

28. Find the shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$.

29. Maximise and minimise $Z=x+2y$ subject to the constraints

$$x+2y\ \ge\ 100$$

$$2x-y\ \le\ 0$$

$$2x+y\ \le\ 200$$

$$x,\ y\ \ge\ 0$$

Solve the above LPP graphically.

30. If $A=\begin{bmatrix}2 & 3\\ 1 & -4\end{bmatrix}$, $B=\begin{bmatrix}1 & -2\\ -1 & 3\end{bmatrix}$, verify that $(AB)^{-1}=B^{-1}A^{-1}$.

31. Find $\int\frac{(2x-5)\,e^{2x}}{(2x-3)^3}dx$.

OR

Evaluate $\int\frac{x\sin^{-1}x}{\sqrt{1-x^2}}dx$.

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SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. If $y = \tan^{-1}\left(\frac{\sqrt{1+x^2}+\sqrt{1-x^2}}{\sqrt{1+x^2}-\sqrt{1-x^2}}\right)$, $x^2 \leq 1$, then find dy/dx.

OR

Find whether the following function is differentiable at $x = 1$ and $x = 2$ or not.

$$f(x) = \begin{cases} x, & x < 1 \\ 2 - x, & 1 \leq x \leq 2 \\ -2 + 3x - x^2, & x > 2 \end{cases}$$

33. Using vectors, find the area of the ΔABC, whose vertices are $A(1,2,3)$, $B(2,-1,4)$ and $C(4,5,-1)$.

OR

If lines $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-k}{2} = \frac{z}{1}$ intersect, then find the value of k and hence, find the equation of the plane containing these lines.

34. Using integration, find the area of the region bounded by the triangle whose vertices are $(-1,0)$, $(1,3)$ and $(3,2)$.

OR

Using integration, find the area of the triangular region whose have the equation $y = 2x + 1$, $y = 3x + 1$ and $x = 4$.

35. Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by $(a,b)R(c,d)$ if $ad(b+c) = bc(a+d)$. Show that R is an equivalence relation.

OR

Show that the function f in $A = R - \left\{\frac{2}{3}\right\}$ defined as $f(x) = \frac{4x+3}{6x-4}$ is one-one and onto.

SECTION - E

Case study based questions are compulsory.

36. ICAR-Indian Agricultural Research Institute is an autonomous body responsible for co-ordinating agricultural education and research in India. It reports to the Department of Agricultural Research and Education, Ministry of Agriculture. The Union Minister of Agriculture serves as its president. It is the largest network of agricultural research and education institutes in the world.



ICAR grows vegetables and grades each one as either good or bad for its taste, good or bad for its size, and good or bad for its appearance. Overall 78% of the vegetables have a good taste. However, only 69% of the vegetables have both a good taste and a good size. Also, 5% of the vegetables have both a good taste and a good appearance, but a bad size. Finally, 84% of the vegetables have either a good size or a good appearance.

(i) If a vegetable has a good taste, what is the probability that it also has a good size?

(ii) If a vegetable has a bad size and a bad appearance, what is the probability that it has a good taste?

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37. Pastry is a dough of flour, water and shortening that may be savoury or sweetened. Sweetened pastries are often described as bakers' confectionery. The word "pastries" suggests many kinds of baked products made from ingredients such as flour, sugar, milk, butter, shortening, baking powder, and eggs.



The Sunrise Bakery Pvt Ltd produces three basic pastry mixes A, B and C. In the past the mix of ingredients has shown in the following matrix:

$$\begin{array}{cc} & \begin{array}{ccc}\text{Flour} & \text{Fat} & \text{Sugar}\end{array} \\ \text{Type}\begin{array}{c}A\\B\\C\end{array} & \begin{bmatrix}5 & 1 & 1\\ 6.5 & 2.5 & 0.5\\ 4.5 & 3 & 2\end{bmatrix}\end{array} \quad \text{(All quantities in kg)}$$

Due to changes in the consumer's tastes it has been decided to change the mixes using the following amendment matrix:

$$\begin{array}{cc} & \begin{array}{ccc}\text{Flour} & \text{Fat} & \text{Sugar}\end{array} \\ \text{Type}\begin{array}{c}A\\B\\C\end{array} & \begin{bmatrix}0 & 1 & 0\\ -0.5 & 0.5 & 0.5\\ 0.5 & 0 & 0\end{bmatrix}\end{array}$$

Using matrix algebra you are required to calculate:

(i) The matrix for the new mix:

(ii) The production requirement to meet an order for 50 units of type A, 30 units of type B and 20 units of type C of the new mix;

(iii) The amount of each type that must be made to totally use up 370 kg of flour, 170 kg of fat and 80 kg of sugar that are at present in the stores.

38. Brine is a high-concentration solution of salt in water. In diverse contexts, brine may refer to the salt solutions ranging from about 3.5% up to about 26%. Brine forms naturally due to evaporation of ground saline water but it is also generated in the mining of sodium chloride.



A tank initially contains 10 gallons of pure water. Brine containing 3 pounds of salt per gallon flows into the tank at a rate of 2 gallons per minute, and the well-stirred mixture flows out of the tank at the same rate.

(i) How much salt is present at the end of 10 minutes?

(ii) How much salt is present in the long run?

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